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C01 ASSESSMENT TOOL1 -APPLICATION BASED QUESTION

DSA0308-NLP

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1.Resume information extraction using regular expression.

Explanation:

The resume information extraction system is developed using python and regular expression to automatically extract candidate details.

1.Extract Candidate Name:

Identity and extract the candidate’s name from the resume text

2.Identify email address:

Uses regular expressions to find the candidate email address

3.Extract mobile number:

Delects and extract the candidate’s mobile number from the resume.

4.Delect Technical skills:

Identifies technical skills such as python, java, Sql, Machine learning and Nlp

5.Extract years of experience:

Find the candidate work experience in the years from the resume

6.Generate structured summary:

Organize the extracted details, including name, email, mobile number, skill, and experience, into a clear candidate profile

7.Check Eligibility criteria:

Shortlists candidate who have at least 2 year of experience and possess python skills. Candidate who do not satisfy both condition are marked as not eligible

Result:

The system extract candidate information automatically, generates structured profile and identifies eligible candidate, reducing manual work and saving time for the HR department

2.Product search system using Regular expressions

Explanation:

The product search system is developed using python and Regular expressions to help customers find product easily in an e-commerce application

1.Exact keyword search:

Searches for a product that exactly matches the given keyword matching

2.Prefix- based search:

Find product whose names start with the specified keyword.

3. Suffix-Based Search:

Finds products whose names end with the specified keyword.

4. Partial Keyword Search:

Finds products containing the given keyword anywhere in their names.

5. Case-insensitive search:

Searches for products without considering uppercase or lowercase letters.

6. Display matching products:

Displays all product names that match the given search conditions.

7. Generate Search Report:

Counts and displays the total number of matching products for each type of search.

Result:

The system performs flexible product searches using Regular Expressions, displays matching product names, and generates a report showing the total number of matching products for each search.

3.University Registration System Using Regular Expressions

Explanation:

1. A university registration system is developed to verify student information before allowing students to enroll in courses.

2. The system collects important student details, including the register number, institutional email address, course code, semester information, and mobile number.

3. Register Number Validation: The system checks whether the student's register number follows the format specified by the university.

4. Institutional email validation:It verifies whether the student's email address is correctly formatted and belongs to the university's official email domain.

5. Course code validation: The system checks whether the entered course code contains the required letters and numbers according to the university's rules.

6. Semester validation: It verifies whether the entered semester is within the permitted range, such as semesters 1 to 8.

7. Mobile number validation: The system checks whether the mobile number contains the required number of digits and follows the specified format.

8. Validation messages: After checking each field, the system displays an appropriate message indicating whether the entered information is valid or invalid.

9. Final registration status:The system examines the validation results of all required fields. If every field is valid, the registration is marked as successful. If any field is invalid, the registration is marked as unsuccessful.

10. Report generation: Finally, the system displays a registration report containing the validation status of each field and the overall registration result.

Result

The university registration validation system verifies student details using Regular Expressions, identifies invalid information, displays appropriate validation messages, and determines whether the registration is successful.